

Disentangling Fact from Sentiment: A Dynamic Conflict-Consensus Framework for Multimodal Fake News Detection

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Abstract—Prevalent multimodal fake news detection relies on consistency-based fusion, yet this paradigm fundamentally misinterprets critical cross-modal discrepancies as noise, leading to over-smoothing, which dilutes critical evidence of fabrication. Mainstream consistency-based fusion inherently minimizes feature discrepancies to align modalities, yet this approach fundamentally fails because it inadvertently smooths out the subtle cross-modal contradictions that serve as the primary evidence of fabrication. To address this, we propose the Dynamic Conflict-Consensus Framework (DCCF), an inconsistency-seeking paradigm designed to amplify rather than suppress contradictions. First, DCCF decouples inputs into independent Fact and Sentiment spaces to distinguish objective mismatches from emotional dissonance. Second, we employ physics-inspired feature dynamics to iteratively polarize these representations, actively extracting maximally informative conflicts. Finally, a conflict-consensus mechanism standardizes these local discrepancies against the global context for robust deliberative judgment. Extensive experiments conducted on three real world datasets demonstrate that DCCF consistently outperforms state-of-the-art baselines, achieving an average accuracy improvement of 3.52% .

Index Terms—Fact Disentanglement, Fake News Detection, Inconsistency Detection, Multimodal Learning, Semantic Disentanglement

I. INTRODUCTION

Digital platforms amplify misinformation via deceptive multimodal content. While automated defense is critical, mainstream approaches predominately employ consistency-based fusion to align cross-modal features. However, we argue this premise is fundamentally flawed: the essence of fake news lies in inconsistency, subtle clashes between visual and textual evidence. By prioritizing alignment and treating discrepancies as noise, existing models inadvertently dilute the conflicting

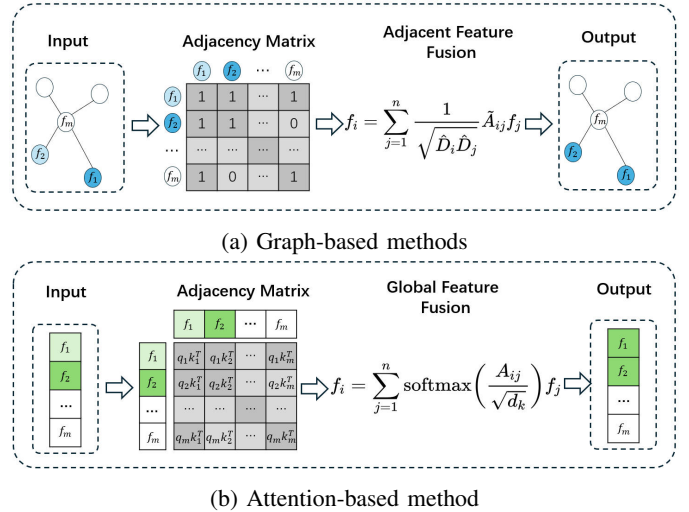


Fig. 1: Schematic of inconsistency distortion, where f_1 and f_2 denote a pair of inconsistent features (e.g., conflicting text and image). (a) Graph-based methods blur this specific conflict by averaging neighbor features via edges. (b) Attention-based methods dilute the inconsistency by aggregating global features via weights, resulting in homogenized outputs.

signals that serve as the primary evidence of fabrication. Consequently, effective detection demands a paradigm shift: moving from seeking consensus to explicitly modeling and amplifying inconsistency.

Fake news detection methods generally fall into three categories. Unimodal methods analyze single data streams, yet this isolation creates information islands that overlook critical cross-modal inconsistencies. Multimodal fusion approaches, ranging from early concatenation (BMR [1]) to co-attention (SEER [2]), typically adopt consistency-seeking paradigms. These methods inadvertently smooth out vital conflict signals

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by treating discrepancies as noise and conflating objective facts with subjective sentiment. Recently, Large Language Models (LLMs) have been leveraged for their reasoning capabilities in approaches like INSIDE [3] and LIFE [4]. Nevertheless, the inherent focus of these models on semantic alignment hinders their ability to effectively capture and amplify the fine-grained inconsistent evidence essential for robust detection.

Despite innovations, existing methods suffer from fundamental limitations: (1) Semantic Entanglement, where objective content and subjective emotion are treated as a mixed signal, blurring the distinction between what is depicted and how it is described, making it difficult to distinguish factual mismatches from emotional dissonance. We resolve this by explicitly disentangling inputs via multi-task supervision. (2) Inconsistency Attenuation, where existing approaches prioritize alignment, inadvertently treating meaningful contradictions as noise and filtering them out, effectively smoothing out the critical discrepancy signals. In contrast, our framework pioneers an inconsistency-seeking paradigm, deploying a tension field network to explicitly amplify feature repulsions and extract maximally informative conflicts as primary evidence.

To address these limitations, we propose the **Dynamic Conflict-Consensus Framework (DCCF)**. Inspired by physical field theory where tension [5] reflects the intensity of differences, our framework adopts a novel approach that actively searches for inconsistencies. It first separates inputs into factual content and emotional tone using multi-task supervision, guided by YOLO [6] and SenticNet [7], to distinguish objective entities from subjective feelings. A fact sentiment tension field network, modeling dynamic forces between features, then iteratively refines these features to highlight their differences. This process amplifies the contrast to extract the most significant conflicts as primary evidence, while simultaneously summarizing the overall style to serve as a global reference. By evaluating specific local conflicts against this global context, DCCF achieves robust, interpretable predictions. Our main contributions are:

- 1) We propose DCCF, a novel inconsistency-seeking paradigm for multimodal fake news detection (MFND). Unlike consistency-seeking methods that blur critical signals, our framework models feature dynamics to amplify and extract inconsistency as primary evidence.
- 2) We introduce an end-to-end fact-sentiment tension field network that quantifies tension metrics to expose latent inconsistencies. By standardizing extreme conflicts against global consensus, it transforms abstract feature dynamics into interpretable reasoning, pinpointing the exact evidence of fabrication.
- 3) We validate DCCF's effectiveness through extensive experiments on widely used MFND benchmarks. Our scheme shows significant performance gains over state-of-the-art baselines, demonstrating superior reliability and robustness.

II. RELATED WORK

A. Multimodal Fake News Detection

Early unimodal methods [8] were insufficient, leading to multimodal detection. Text-visual fusion often interprets images superficially. This, with isolated text features, creates information islands and weak reasoning, failing cross-modal inconsistency detection. Our multi-stage framework addresses this [9]. We extract diverse factual/sentimental features for dynamic evolution, then extract high level metrics of conflict, consensus, and inconsistency. This focus on evolved relationships, not raw features [10], enables robust multi-view judgment by reasoning about inconsistencies.

B. Multi-Domain Fake News Detection

Multi-domain learning models news data spanning diverse domains [11]. Approaches use hard sharing for domain-specific/cross-domain knowledge or soft sharing, like gating networks or domain memory banks. However, these methods often just adjust view weights via domain embeddings, failing to learn domain invariant/specific information [12]. Concatenating embeddings may fail to account for domain dependencies. Furthermore, these are single modal, struggling with rich visual information [3].

Despite their architectural variations, existing unimodal, fusion-based [2], and LLM-driven [3] methods predominantly rely on a consistency-seeking paradigm [4], which aligns features and inadvertently smooths out the critical discrepancies indicative of deception. In contrast, our DCCF pioneers an inconsistency-seeking paradigm, explicitly modeling and amplifying these cross-modal conflicts to leverage inconsistency as the primary evidence for detection [1].

III. METHODOLOGY

We propose the DCCF framework to detect multimodal fake news by modeling feature dynamics within decoupled semantic spaces. DCCF first disentangles inputs into distinct fact and sentiment spaces via multi-task learning, then employs a tension field network to identify polarization and inconsistency. As shown in Fig. 2, the framework comprises three progressive stages: (1) **Fact-Sentiment Feature Extraction**, (2) **Feature Dynamics Evolution and Conflict-Consensus Metric Extraction**, and (3) **Multi-View Deliberative Judgment**.

A. Fact-Sentiment Feature Extraction

This stage projects input text and images onto specialized fact and sentiment feature spaces. This separation effectively distinguishes objective factual inconsistencies from subjective sentimental conflicts, addressing the limitation where monolithic processing often conflates these distinct signals by anchoring features to visual objects and textual polarity, respectively.

Initial Encoding. Using pretrained BERT [13] and ViT [14], we extract initial embeddings e_T and e_I from the raw text T and image I :

$$e_T = \text{BERT}(T), \quad e_I = \text{ViT}(I) \quad (1)$$

Fact Space Projection. Two independent MLPs project e_T and e_I into a shared fact space, yielding f_T^F and f_I^F . To strictly enforce objectivity and semantic alignment, we introduce an auxiliary task:

1. The image I is fed into a pretrained YOLO [6] to generate pseudo-labels e_Y representing key factual objects (e.g., entities, locations).
2. We train the projection to accurately predict e_Y via a BCE Loss $\mathcal{L}_F$, ensuring the space captures grounded reality:

$$\mathcal{L}_F = \text{BCE}(\text{MLP}_{\text{classifier}}(f_I^F), e_Y) \quad (2)$$

Sentiment Space Projection. Simultaneously, two separate MLPs project e_T and e_I into a sentiment space, producing f_T^E and f_I^E . We orient this space towards subjective sentiment via another auxiliary task to capture high level affective semantics:

1. Text T is processed by a lexicon (e.g., SenticNet) to obtain a sentiment polarity vector e_J .
2. We enforce f_T^E to predict this polarity using MSE Loss $\mathcal{L}_E$:

$$\mathcal{L}_E = \text{MSE}(\text{MLP}_{\text{classifier}}(f_T^E), e_J) \quad (3)$$

The output comprises the fact feature space $S^F = \{f_T^F, f_I^F\}$ and sentiment feature space $S^E = \{f_T^E, f_I^E\}$.

B. Feature Dynamics Evolution and Conflict-Consensus Metric Extraction

A dynamic evolution module (inspired by physical tension field theory) operates on S^F and S^E to amplify inconsistency (conflict) and distill global context (consensus).

Feature Dynamics Evolution. Given input $S^{(0)} = \{f_1^{(0)}, \dots, f_n^{(0)}\}$ of n features, a Dynamic Feature Evolution Unit (DARFU) iterates M times ($t = 0$ to $M - 1$):

Compute tension. Calculate tension $T_{i,j}^{(t)}$ between vectors $f_i^{(t)}, f_j^{(t)} \in S^{(t)}$, representing the potential difference in the semantic field:

$$T_{i,j}^{(t)} = (f_i^{(t)} - f_j^{(t)})^2 \quad (4)$$

Compute weights. Convert high tension to low attraction weight via softmax:

$$W_{i,j}^{(t)} = \text{softmax}_j(-T_{i,j}^{(t)}) \quad (5)$$

Aggregate and transform. Update features via residual weighted sum and non-linear transformation g for state $S^{(t+1)}$:

$$f_i^{(t+1)} = f_i^{(t)} + g\left(\sum_{j=1}^n W_{i,j}^{(t)} f_j^{(t)}\right) \quad (6)$$

Crucially, this iterative process acts as an inconsistency-seeking filter: it clusters semantically consistent neighbors while polarizing inconsistent ones, thereby preventing the over-smoothing of conflicting evidence common in graph-based fusion. The output is the final space $S' = S^{(M)}$ and tension matrix $T' = T^{(M-1)}$.

Conflict-Consensus Metric Extraction. From S' and T' , we extract two key metrics to quantify internal contradictions:

1. **Maximally Informative Conflicts.** We identify the pair (f'_i, f'_j) in T' with maximum tension as the key local conflict I_{conflict} :

$$I_{\text{conflict}} = \text{concat}(f'_i, f'_j) \quad \text{where } (i, j) = \arg \max(T') \quad (7)$$

2. **Global Consensus.** We compute the mean of S' as $C_{\text{consensus}}$ to represent the global context/tone:

$$C_{\text{consensus}} = \frac{1}{n} \sum_{i=1}^n f'_i \quad (8)$$

Tone-reference standardization. We concatenate I_{conflict} and $C_{\text{consensus}}$ into an MLP g_{std} . This Tone-reference inconsistency standardization uses consensus to standardize conflict, ensuring that the magnitude of discrepancy is evaluated relative to the document's specific semantic baseline rather than in isolation:

$$V = g_{\text{std}}(\text{concat}(I_{\text{conflict}}, C_{\text{consensus}})) \quad (9)$$

The outputs are refined fact (V^F) and sentiment (V^E) inconsistency vectors.

C. Multi-View Deliberative Judgment

This stage deliberates on the standardized inconsistencies from both semantic subspaces to form a final judgment.

View fusion. We concatenate V^F and V^E into a final representation V_{final} , encapsulating complementary dual-view inconsistency:

$$V_{\text{final}} = \text{concat}(V^F, V^E) \quad (10)$$

Final classification. V_{final} is fed into a classifier (e.g., MLP with sigmoid) to predict the probability $\hat{y}$ that the news is Fake.

D. Loss Function

The total loss $\mathcal{L}_{\text{total}}$ combines the prediction BCE loss $\mathcal{L}_{\text{final}}$ with auxiliary losses $\mathcal{L}_F$ and $\mathcal{L}_E$, balanced by λ_F and λ_E :

$$\mathcal{L}_{\text{total}} = (1 - \lambda_F - \lambda_E)\mathcal{L}_{\text{final}} + \lambda_F\mathcal{L}_F + \lambda_E\mathcal{L}_E \quad (11)$$

Joint optimization ensures the model accurately captures visual objective cues, textual sentiment nuances, and critical inconsistencies, yielding robust detection performance.

IV. EXPERIMENTS

A. Experimental Settings

We validate DCCF with experiments detailing our datasets, baselines, and implementation.

Datasets. We use three benchmarks: Weibo [11], Weibo21 [15], and GossipCop [16], following established protocols. Weibo [11] includes 7,532 training (3,749 real/3,783 fake) and 1,996 test (996 real/1,000 fake) articles. Weibo21 [15] has 9,127 total articles (4,640 real/4,487 fake). GossipCop [16] provides 10,010 training (7,974 real/2,036 fake) and 2,830 testing (2,285 real/545 fake) instances.

Baselines. We benchmark against three categories: (1) Unimodal methods (MVAN [10], SpotFake [8]). (2) Cross-domain

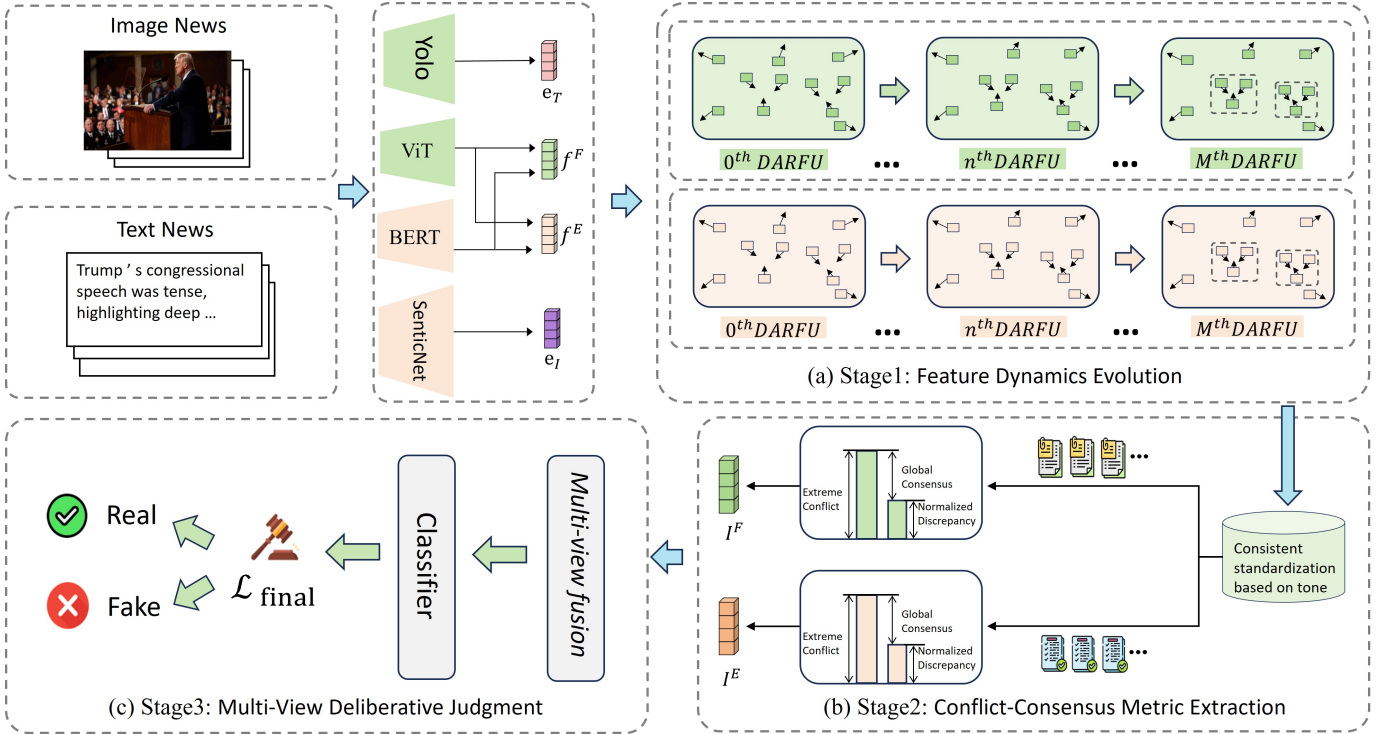


Fig. 2: The DCCF framework: (a) Fact-Sentiment Feature Extraction projects features into Fact (S^F) and Sentiment (S^E) spaces; (b) Feature Dynamics Evolution refines features through DARFU blocks to compute conflict/consensus; (c) Multi-View Deliberative Judgment fuses metrics for the final decision.

generalization (EANN [11], FND-CLIP [12], MIMoE-FND [16], KEN [17]). (3) LLM distillation (GLPN-LLM [18], INSIDE [3], LIFE [4]).

Implementation Details. Visual features used a pretrained MAE [19] with 224×224 images. Text used bert-base-chinese [13] (Weibo/Weibo21) and bert-base-uncased [13] (Gossip-Cop), truncated to 197 tokens. Features were aligned using CLIP [20]. The auxiliary loss coefficients λ_F and λ_E were both set to 0.075. The model used PyTorch, trained on one NVIDIA RTX 4090 GPU for 50 epochs with early stopping.

B. Overall Performance

To validate DCCF’s superiority, we compare it against 11 baselines on three datasets (Table I). From the results, we draw these key observations:

(O1): DCCF consistently achieves state-of-the-art performance across diverse benchmarks. On Weibo, it surpasses the strongest baseline (LIFE) by 1.1% in accuracy and 2.7% in F1-Fake, highlighting its capability in detecting standard multimodal inconsistencies.

(O2): On the recent Weibo-21 dataset, DCCF maintains a competitive edge. Despite a narrower margin against the top baseline (MIMoE-FND), our model secures the best results across all four metrics. This confirms that our dynamic conflict-seeking paradigm generalizes effectively to varying data distributions.

(O3): DCCF demonstrates superior robustness on the imbalanced GossipCop (80% real). Unlike baselines that

trade off F1 scores, DCCF achieves the highest accuracy (0.904) and balanced performance (F1-Fake: 0.723, F1-Real: 0.946), effectively mitigating class imbalance pitfalls.

(O4): The results validate our architectural hypothesis. Disentangling features into fact/sentiment spaces and amplifying inconsistency via the tension field network proves more effective than both traditional fusion (e.g., BMR [1], SEER [2]) and LLM-distillation approaches (e.g., INSIDE [3], LIFE [4]).

C. Ablation Study

To understand DCCF’s core components, we ran an ablation study (Table II).

Effect of Multi-View Deliberative Judgment. Removing the Fact View (w/o Fact View) or Sentiment View (w/o Sentiment View) degraded performance, especially the former. This confirms integrating inconsistency signals from factual and sentimental spaces, validating our multi-view architecture.

Effect of Feature Disentanglement Components. Removing auxiliary losses (w/o $\mathcal{L}_{Yolo}$) or (w/o $\mathcal{L}_{SenticNet}$) hurt performance, especially $\mathcal{L}_{Yolo}$, verifying the importance of multi-task learning. Removing both (w/o $\mathcal{L}_{Double}$) caused a severe drop, proving this separation is critical.

Effect of Tension Field Network and Metric Extraction. Removing the Feature Dynamics Evolution stage (w/o Evolution) caused a severe performance drop (e.g., 5.2% on Weibo). Removing the tension-to-weight mechanism (w/o Tension Weighting) also degraded performance. Furthermore,

TABLE I: Performance comparison of DCCF against state-of-the-art methods on Weibo, Weibo-21, and GossipCop datasets. The row “Improv.” indicates the improvement of our model compared to the best-performing baseline.

Method	Weibo				Weibo-21				GossipCop			
	Acc	F1-Fake	F1-Real	AUC	Acc	F1-Fake	F1-Real	AUC	Acc	F1-Fake	F1-Real	AUC
<i>Multimodal Multi-domain</i>												
EANN	0.827	0.829	0.825	0.873	0.870	0.862	0.875	0.894	0.864	0.594	0.920	0.852
FND-CLIP	0.907	0.908	0.907	0.953	0.943	0.940	0.946	0.962	0.880	0.638	0.928	0.871
MIMoE-FND	0.928	0.928	0.928	0.972	0.956	0.955	0.957	0.977	0.895	0.698	0.938	0.879
KEN	0.935	0.935	0.934	0.967	0.935	0.937	0.932	0.971	0.881	0.646	0.928	0.873
<i>Multimodal Single-domain</i>												
CAFE	0.840	0.842	0.837	0.892	0.882	0.885	0.876	0.909	0.867	0.587	0.921	0.852
BMR	0.918	0.914	0.904	0.954	0.929	0.927	0.925	0.962	0.895	0.691	0.876	0.881
RaCMC	0.915	0.917	0.914	0.921	0.942	0.938	0.943	0.962	0.879	0.641	0.927	0.838
SEER	0.929	0.928	0.939	0.934	0.932	0.927	0.925	0.960	0.893	0.673	0.871	0.875
<i>LLM-based Reasoning</i>												
INSIDE	0.881	0.684	0.932	0.910	0.896	0.816	0.832	0.871	0.900	0.707	0.934	0.881
GLPN-LLM	0.920	0.939	0.921	0.954	0.925	0.937	0.924	0.959	0.890	0.682	0.933	0.864
LIFE	0.924	0.940	0.921	0.958	0.928	0.935	0.934	0.975	0.864	0.701	0.866	0.855
Ours												
DCCF	0.951	0.951	0.954	0.978	0.957	0.957	0.965	0.982	0.904	0.723	0.946	0.889
Improv.	1.71%↑	1.17%↑	1.60%↑	0.62%↑	0.10%↑	0.21%↑	0.84%↑	0.51%↑	0.44%↑	2.26%↑	0.85%↑	0.91%↑

TABLE II: Ablation study results showing performance drops for different model variants.

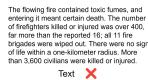
Model	Weibo			Weibo-21			GossipCop		
	Acc	F1-Fake	F1-Real	Acc	F1-Fake	F1-Real	Acc	F1-Fake	F1-Real
DCCF	0.951	0.951	0.954	0.957	0.957	0.965	0.904	0.723	0.946
<i>Multi-task Feature Learning</i>									
- w/o $\mathcal{L}_{Yolo}$	0.925 $\downarrow 2.7\%$	0.926 $\downarrow 2.6\%$	0.922 $\downarrow 3.4\%$	0.933 $\downarrow 2.5\%$	0.930 $\downarrow 2.8\%$	0.934 $\downarrow 3.2\%$	0.885 $\downarrow 2.1\%$	0.679 $\downarrow 6.1\%$	0.922 $\downarrow 2.5\%$
- w/o $\mathcal{L}_{SenticNet}$	0.927 $\downarrow 2.5\%$	0.920 $\downarrow 3.3\%$	0.927 $\downarrow 2.8\%$	0.928 $\downarrow 3.0\%$	0.932 $\downarrow 2.6\%$	0.939 $\downarrow 2.7\%$	0.880 $\downarrow 2.7\%$	0.678 $\downarrow 6.2\%$	0.919 $\downarrow 2.9\%$
- w/o $\mathcal{L}_{Both}$	0.922 $\downarrow 3.1\%$	0.912 $\downarrow 4.1\%$	0.908 $\downarrow 4.8\%$	0.941 $\downarrow 1.7\%$	0.917 $\downarrow 4.2\%$	0.920 $\downarrow 4.7\%$	0.885 $\downarrow 2.1\%$	0.673 $\downarrow 6.9\%$	0.909 $\downarrow 3.9\%$
<i>DARFU</i>									
- w/o Evolution	0.931 $\downarrow 2.1\%$	0.928 $\downarrow 2.4\%$	0.924 $\downarrow 3.1\%$	0.943 $\downarrow 1.5\%$	0.936 $\downarrow 2.2\%$	0.939 $\downarrow 2.7\%$	0.879 $\downarrow 2.8\%$	0.692 $\downarrow 4.3\%$	0.928 $\downarrow 1.9\%$
- w/o Tension Weighting	0.934 $\downarrow 1.8\%$	0.924 $\downarrow 2.8\%$	0.922 $\downarrow 3.4\%$	0.947 $\downarrow 1.0\%$	0.934 $\downarrow 2.4\%$	0.935 $\downarrow 3.1\%$	0.870 $\downarrow 3.8\%$	0.686 $\downarrow 5.1\%$	0.922 $\downarrow 2.5\%$
<i>Conflict-Consensus Metric Extraction</i>									
- w/o Maximally Informative Conflicts	0.915 $\downarrow 3.8\%$	0.918 $\downarrow 4.3\%$	0.920 $\downarrow 3.6\%$	0.911 $\downarrow 4.8\%$	0.923 $\downarrow 3.6\%$	0.924 $\downarrow 4.3\%$	0.863 $\downarrow 4.5\%$	0.683 $\downarrow 5.5\%$	0.921 $\downarrow 2.6\%$
- w/o Global Consensus	0.900 $\downarrow 5.4\%$	0.912 $\downarrow 4.1\%$	0.913 $\downarrow 4.3\%$	0.919 $\downarrow 4.0\%$	0.927 $\downarrow 3.1\%$	0.932 $\downarrow 3.4\%$	0.858 $\downarrow 5.1\%$	0.677 $\downarrow 6.4\%$	0.922 $\downarrow 2.5\%$
<i>Multi-View Deliberative Judgment</i>									
- w/o Fact View	0.913 $\downarrow 4.0\%$	0.918 $\downarrow 3.5\%$	0.922 $\downarrow 3.4\%$	0.920 $\downarrow 3.9\%$	0.925 $\downarrow 3.3\%$	0.930 $\downarrow 3.6\%$	0.861 $\downarrow 4.8\%$	0.671 $\downarrow 7.2\%$	0.908 $\downarrow 4.0\%$
- w/o Sentiment View	0.919 $\downarrow 3.4\%$	0.916 $\downarrow 3.7\%$	0.924 $\downarrow 3.1\%$	0.933 $\downarrow 2.5\%$	0.926 $\downarrow 3.2\%$	0.931 $\downarrow 3.5\%$	0.887 $\downarrow 1.9\%$	0.684 $\downarrow 5.4\%$	0.926 $\downarrow 2.1\%$

removing metric components (w/o Global Consensus or w/o Maximally Informative Conflicts) caused degradation, demonstrating the effectiveness of DCCF components.

D. Parameter Sensitivity Analysis

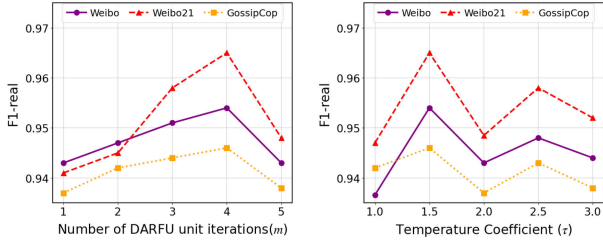
We analyzed parameter sensitivity on three datasets, focusing on four key parameters: DARFU iterations (m), temperature coefficient (τ), and the fact auxiliary loss coefficient (λ_F) and the sentiment auxiliary loss coefficient (λ_E). As shown in the figures, the model is robust. Performance forms a bell shaped curve, peaking at $m=4$, $\tau=1.5$, and a 7.5% loss weight for both fields. Performance gracefully declines from these optimal points but remains high, demonstrating the model is effective across various configurations and not overly sensitive.

TABLE III: DCCF explainability case study on two challenging examples from the GossipCop test set. The calibration allocation abbreviations denote: FT (Fact-Text), FI (Fact-Image), ST (Sentiment-Text), and SI (Sentiment-Image).

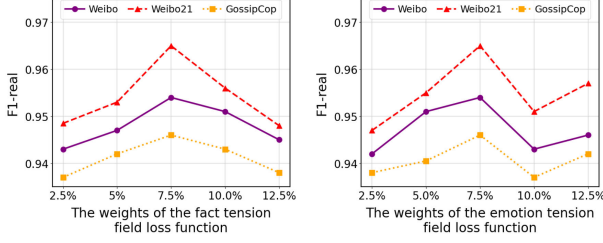
	News Posts	Fake Type	Ground Truth	Calibration Allocation	DCCF	MIMoE-FND
Case 1	 The flowing fire contained toxic fumes, and... Text $\times$	Text Fabrication	Fake news	FT (41.8%)	Fake news	Fake news
				ST (58.2%)		
Case 2	 Yoshi Smith suddenly stopped the music midway... Image $\times$	Image-Text Mismatch	Fake news	FI (20.7%)	Fake news	Real news
				SI (32.4%)		

E. Case Study: Model Explainability

We investigate DCCF’s interpretability with a case study of two challenging GossipCop instances. This illustrates how



(a) Effects of the number of DARFU unit iterations (m) and the temperature coefficient (τ).



(b) Effects of the fact auxiliary loss coefficient ($\mathcal{L}_F$) and the sentiment auxiliary loss coefficient ($\mathcal{L}_E$).

Fig. 3: Analysis of hyperparameter sensitivity. This figure shows the impact of four different hyperparameters on the model’s F1-real score across three datasets. (a) Effects of the number of DARFU unit iterations (m) and the Temperature Coefficient (τ). (b) Effects of the fact auxiliary loss coefficient ($\mathcal{L}_F$) and the sentiment auxiliary loss coefficient ($\mathcal{L}_E$).

DCCF acquires and distills deep reasoning into its text and image representations.

The first case is a Text Fabrication. Both DCCF and the baseline MIMoE-FND [16] correctly classified it.

The second case is an Image-Text mismatch. DCCF correctly predicted Fake news, while the baseline MIMoE-FND [16] failed, misclassifying it as Real news.

F. T-SNE Visualizations

Fig. 4 shows T-SNE visualizations of features from DCCF, MIMoE-FND [16], and KEN [17] on the Weibo and Weibo21 test sets. Compared to the baselines, DCCF produces fewer fake news outliers and less overlap between real and fake news embeddings, confirming its superior performance.

On Weibo21, DCCF’s features form multiple, clearly separated subclusters, unlike the single clusters on Weibo. This suggests Weibo21 has varying topics and that DCCF not only distinguishes authenticity but also captures deep, event-level semantic information, spatially distinguishing different events.

V. CONCLUSION

In this paper, we propose the DCCF, a novel inconsistency-seeking paradigm. DCCF initially employs fact sentiment feature extraction guided by multi-task supervision to decouple semantic spaces. Subsequently, the fact sentiment tension field network iteratively models feature dynamics to polarize representations, distilling interpretable maximally informative conflicts and global consensus metrics. Finally, the Multi-View Deliberative Judgment fuses these standardized indicators for

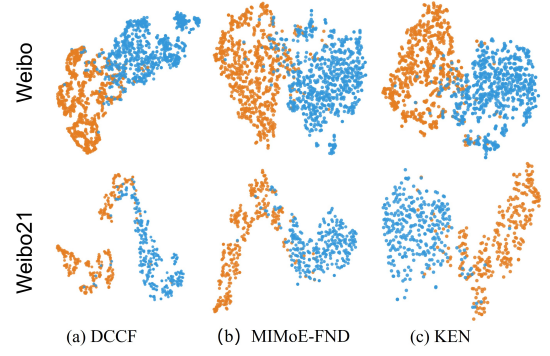


Fig. 4: T-SNE visualization of test set features. Same color dots indicate the same label.

robust detection. Extensive experiments validate DCCF’s superiority. A primary limitation is the framework’s reliance on the quality of auxiliary pseudo-labels, where upstream noise may propagate to the decoupled spaces. Future work will explore integrating Large Language Models to enhance the robustness of these semantic constraints.

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